Extending BlockSim with Energy and Carbon Footprint Modeling for Sustainable Blockchain Evaluation

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Abstract –This paper extends BlockSim by introducing energy-consumption and carbon-footprint metrics that are tightly integrated with its event-driven execution model, enabling sustainability-aware evaluation alongside conventional performance metrics. The proposed extension instruments core simulation events to estimate computational and communication energy at both node and network levels, then converts electricity demand to CO_2 emissions using an emission-factor formulation that can be configured to represent different grid carbon intensities. Using the extended simulator, controlled experiments are conducted on Proof-of-Work reference configurations (Bitcoin, and a historical Ethereum-like PoW configuration that does not represent current Ethereum, which has used Proof-of-Stake since The Merge) under varying miner populations and workloads. The framework reports aggregate energy and CO_2, as well as normalized indicators per block and per transaction, supporting reproducible "what-if" analysis without external post-processing and enabling direct comparison of protocol configurations as networks scale.

In this revised version, the framework additionally distinguishes economically driven Proof-of-Work (PoW) energy consumption, in which the expected mining reward and cryptocurrency price are explicit inputs, from validator-count-driven Proof-of-Stake (PoS) energy consumption; it varies the carbon emission factor across grid scenarios; and it is positioned as a scenario-based evaluation tool for preliminary what-if analysis rather than a precise real-world energy estimator.

Keywords: Blockchain Simulation, BlockSim, Consensus Algorithms, Energy Consumption, PoW. Proof of Stake, Carbon Footprint, Energy Modeling, Sustainability.

Introduction

Blockchain technology has become a crucial infrastructure for decentralized applications, facilitating trustless coordination within distributed systems. Following Bitcoin's inception, blockchains have seen widespread adoption in sectors including finance, supply chain management, and digital governance. Nevertheless, the swift expansion of blockchain networks has prompted considerable apprehension regarding their energy consumption and environmental consequences, especially concerning consensus mechanisms reliant on computationally intensive processes like Proof of Work (PoW) [1].

Numerous studies have shown that large-scale blockchain systems can consume energy levels comparable to those of entire nations, leading to significant carbon dioxide (CO_2) emissions [2], [3]. These environmental issues have spurred the development of more energy-efficient consensus protocols and the necessity for robust evaluation tools capable of assessing sustainability alongside conventional performance and security metrics.

In this context, analysis using simulations is crucial. It allows for controlled experiments, avoiding the costs and risks associated with using real blockchain networks.

BlockSim, developed by Alharby and van Moorsel, is a prominent example of a blockchain simulator, distinguished by its adaptability and layered design. This architectural framework facilitates the examination of design compromises spanning network, consensus, and incentive layers via discrete-event simulation [4]. BlockSim offers functionality for assessing a range of system-level metrics, such as latency, throughput, and fork rates, and has been extensively employed to analyze blockchain performance across different configurations. Nevertheless, like numerous blockchain simulators, BlockSim does not inherently model energy consumption or carbon emissions, which limits its applicability in research centered on sustainability.

Concurrently, energy modeling has been effectively incorporated into versatile network simulators like ns-3, thereby illustrating the capacity of discrete-event simulation to accurately represent power and energy behaviors within distributed systems [5]. This observation implies that analogous modeling methodologies could be applied to blockchain simulators.

This study expands upon BlockSim by incorporating energy consumption and carbon footprint metrics, which are closely integrated with its event-driven execution model. This enhancement facilitates a quantitative assessment of blockchain protocols, considering both performance and environmental impacts, thus furnishing researchers and protocol designers with a comprehensive framework for the sustainable analysis of blockchain systems.

It is important to clarify the energy characteristics of the consensus mechanisms considered. Proof-of-Work (PoW) energy consumption is economically driven: a rational miner invests in electricity up to the fiat value of the expected block reward, so total PoW energy depends primarily on the cryptocurrency price, the block subsidy, transaction fees, the electricity price, mining hardware efficiency, and the difficulty/hashrate equilibrium, rather than on the raw number of miners [16], [4]. Proof-of-Stake (PoS), by contrast, secures the network through stake and its energy is dominated by the number of validators kept online and their hardware power, essentially independent of coin price [19]. Notably, Ethereum transitioned from PoW to PoS at The Merge (September 2022), reducing its consensus energy by roughly 99.95% [21]; accordingly, current Ethereum is not a PoW system, and any PoW Ethereum configuration in this paper is labelled as a historical Ethereum-like PoW configuration only.

Background and Related Work

Blockchain systems are often evaluated using analytical models, empirical measurements, or simulation-based approaches. Simulation, in particular, has gained prominence due to its capacity to effectively model complex interactions among networking, consensus, and incentive mechanisms within controlled and reproducible settings. Consequently, as blockchain protocols evolve, simulators have become essential tools for examining scalability, security, and performance trade-offs, thus eliminating the necessity for costly real-world infrastructure implementations.

Numerous blockchain simulators have been created in existing research. For example, SimBlock focuses on how blocks spread and how forks behave in large peer-to-peer networks. This makes it useful for studying network delays and their impact on the security of consensus mechanisms [6].

JABS builds upon this research by incorporating support for diverse consensus mechanisms and accommodating extensive network scales, thereby prioritizing both performance and fault tolerance [7]. In contrast, other simulators, including those developed on the ns-3 platform, offer high-precision network modeling; however, they necessitate considerable resources to integrate blockchain logic and consensus processes [8].

BlockSim, as described by Alharby and van Moorsel, utilizes a layered discrete-event simulation model, which differentiates among network, consensus, and incentive components [4]. This architectural framework allows for adaptable examination of protocol design alternatives, and it has been employed in various studies to evaluate throughput, latency, and fork rates. Moreover, BlockSim's modular structure makes it particularly well-suited for extensions and experimentation, thereby positioning it as a valuable foundation for future enhancements.

Despite these advancements, blockchain simulation frameworks still largely overlook energy consumption and environmental effects. Although many studies have quantified the energy use and carbon footprint of blockchain networks, particularly Bitcoin and Ethereum [2], [3], using real-world data and statistical models, these analyses are often retrospective and lack the flexibility needed for "what-if" experiments.

General-purpose simulators, like ns-3, include energy frameworks that can model power consumption in networked systems [9]; however, most blockchain-specific simulators don't have these features built-in.

Recent investigations have underscored the necessity for evaluation instruments that incorporate sustainability considerations, thereby integrating performance, security, and environmental metrics within a unified framework [3]. Nevertheless, existing methodologies frequently depend on external post-processing or imprecise approximations, rather than employing tightly integrated simulation models. This deficiency underscores the imperative to augment blockchain simulators, including BlockSim, with explicit modeling of energy consumption and carbon emissions, thus facilitating systematic and comparative sustainability assessments throughout the protocol design process.

Energy, Permissionlessness, and Sybil Resistance

The energy gap between PoW and PoS is tied to how each mechanism resists Sybil attacks in a permissionless setting. PoW achieves Sybil resistance by binding influence to an external, costly resource (computation and therefore electricity), so PoW is not “free”: its permissionlessness is purchased through continuous resource expenditure, which is precisely why its energy use is large and coupled to coin price [16], [4]. PoS instead binds influence to internal economic stake and validator participation, removing most computational waste but introducing different trust and economic assumptions [19]. Platt, Platt and McBurney formalise these tensions as a Sybil-attack vulnerability trilemma among permissionlessness, Sybil resistance, and freeness [20], and De Vries documents how Ethereum’s move to PoS illustrates a sustainability path for the wider ecosystem [21]. These distinctions motivate treating PoW and PoS as two separate energy models in the remainder of the paper.

Overview of BlockSim (Alharby & van Moorsel)

BlockSim is a discrete-event simulation framework that facilitates the systematic assessment of blockchain systems by abstractly modeling their fundamental architectural layers, while also allowing for extensibility. Developed by Alharby and van Moorsel, BlockSim was conceived to overcome the constraints inherent in analytical and empirical evaluation techniques when examining intricate blockchain design trade-offs across diverse configurations [4].

The simulator's architecture is structured in layers, dividing the blockchain system into three main components: the network layer, the consensus layer, and the incentive layer. The network layer simulates message propagation delays and peer connectivity, which allows for the analysis of latency and communication overhead.

The consensus layer, which includes the processes of block creation, validation, and fork resolution, allows researchers to study how protocol parameters, such as block intervals and the distribution of mining power, affect the system.

The incentive layer models economic aspects, including rewards and penalties, providing insight into miner behavior and system stability [4].

BlockSim, constructed using Python, operates within a discrete-event framework, wherein system dynamics unfold through pre-scheduled events encompassing block creation, message dissemination, and chain modifications. This methodology facilitates repeatable experimentation and precise manipulation of simulation variables, while simultaneously providing the necessary abstraction to accommodate extensive scenario analysis. Prior research has demonstrated BlockSim's efficacy in assessing throughput, block propagation latency, and stale block frequencies across diverse blockchain architectures [4], [10].

A significant advantage of BlockSim is its inherent modularity and capacity for extension. The framework is deliberately structured to empower researchers to modify or augment internal models, encompassing node functionality, transaction processing, and consensus mechanisms, without necessitating a complete overhaul of the simulator.

BlockSim's design makes it particularly well-suited for "what-if" analyses and comparative studies of different protocol designs [10]. While it is versatile, BlockSim mainly focuses on performance, security, and incentive-related metrics. The original framework doesn't explicitly model environmental factors like energy use and carbon emissions. Given the growing importance of sustainable blockchain design and the proven ability to integrate energy models into discrete-event simulators [9], BlockSim offers a solid foundation for extensions that include energy and carbon-aware evaluations while preserving its original architectural principles..

Energy Consumption Modeling Extension

Energy consumption in blockchain systems is primarily driven by computational operations associated with consensus execution and communication activities related to block and transaction propagation. To support sustainability-oriented analysis, this work extends BlockSim by integrating an explicit energy consumption model into its discrete-event simulation framework.

Following established practices in energy-aware modeling of distributed systems, energy usage is expressed as a function of discrete events and their associated resource costs [11], [9]. In the proposed extension, each node i maintains a cumulative energy counter E_i, which is incremented whenever a computation or communication event is executed.

A. Computational Energy Model

Computational energy consumption is modeled based on consensus-related operations such as block generation and validation. For a node i, the computational energy consumed during a simulation interval is defined as:

$$E_i^{(comp)}=\sum_{k=1}^{N_i^{(comp)}} e_{(comp)}^{(k)} \quad (1)$$

where $N_i^{(comp)}$ denotes the number of computational events executed by node i, and $e_{(comp)}^{(k)}$ represents the energy cost of the k-th computational event.

For Proof of Work–based mechanisms, $e_{(comp)}$is proportional to the number of hash attempts:

$$e_{(comp)}=\alpha\cdot H_i \quad (2)$$

where H_i is the number of hash computations performed by node i, and α is the energy cost per hash operation, consistent with prior empirical analyses of blockchain energy usage [2]. For less computation-intensive consensus mechanisms, fixed or lower per-event energy coefficients are applied, reflecting reduced processing requirements [12].

B. Communication Energy Model

Communication energy consumption accounts for message transmission and reception events in the network layer. The communication energy for node i is defined as:

$$E_i^{(comm)}=N_i^{(tx)}\cdot e_{tx}+N_i^{(rx)}\cdot e_{rx} \quad (3)$$

where $N_i^{(tx)}$ and $N_i^{(rx)}$ represent the number of transmitted and received messages, respectively, and e_{tx}, e_{rx} denote the energy costs of transmission and reception. This formulation aligns with packet-level energy models commonly used in network simulators such as ns-3 [9].

Although defined here, communication energy was not analysed in the previous version. In this revision it is instrumented with explicit transmit/receive counters for block and transaction messages and analysed in Section (Results) as a function of transaction rate, block size, and peer connectivity, and reported separately from consensus energy.

C. Total Energy Consumption

The total energy consumption per node and for the entire network is computed as:

$$E_i=E_i^{(comp)}+E_i^{(comm)},E_{total}=\sum_{i=1}^{N} E_i \quad (4)$$

This integrated energy model enables BlockSim to quantify energy usage per block, per transaction, and per simulation run, forming a foundation for subsequent carbon footprint analysis.

D. Proof-of-Work Economic Energy Model

Because PoW energy is bounded by mining economics, we add an economic model. The expected per-block reward in fiat is given by Eq. (8), where B_t is the block subsidy, F_t the transaction fees, and P_t the coin price. A rational mining market spends a fraction $\kappa \in [0,1]$ of this reward on electricity, giving the economic energy budget per block in Eq. (9), where $C_{(elec,t)}$ is the electricity price per kWh. When hardware data are available, a technical bound $E_{(technical)}=H_{(block)}\cdot\varepsilon_{hash}$ is derived from the expected hashes per block ($H_{(block)}=D\cdot 2^{32}$) and the per-hash efficiency ε_{hash}, and the network per-block energy is $E_{(PoW,t)}=\min(E_{(technical)}, E_{(budget,t)})$. The miner-level allocation by hashpower share $s_{(i,t)}$ is given by Eq. (10). Thus PoW energy depends on coin price, reward, electricity price, efficiency, and difficulty/hashrate, and only the distribution depends on the number of miners.

$$R_t=(B_t+F_t)\cdot P_t \quad (8)$$

$$E_{(budget,t)}=(\kappa \cdot R_t)/(C_{(elec,t)}) \quad (9)$$

$$E_{(i,t)}=s_{(i,t)} \cdot E_{(PoW,t)} \quad (10)$$

E. Proof-of-Stake Validator Energy Model

For PoS we add a validator-count model: the network energy is given by Eq. (11), where P_v is per-validator power (W), T the horizon (h), $u_v \in [0,1]$ the uptime, V the number of validators, and $E_{(comm)}$ the communication energy. PoS energy therefore scales with validator count and hardware power and is independent of coin price.

$$E_{(PoS)}=(\textstyle\sum_{(v=1..V)} (P_v \cdot T \cdot u_v))/(1000)+E_{(comm)} \quad (11)$$

Carbon Footprint Modeling

While energy consumption provides a direct measure of resource usage, assessing the environmental impact of blockchain systems requires translating energy demand into associated carbon dioxide (CO_2) emissions. Carbon footprint modeling enables sustainability-oriented evaluation by linking simulated energy consumption to greenhouse gas emissions attributable to electricity generation.

In this work, carbon emissions are derived directly from the energy consumption metrics produced by the extended BlockSim framework. Following widely adopted carbon accounting practices, total CO_2 emissions are modeled as a linear function of energy usage and an emission factor that represents the carbon intensity of electricity generation [13], [14].

A. Carbon Emission Model

Let E_i denote the total energy consumption of node i, measured in kilowatt-hours (kWh), as obtained from the energy consumption model. The corresponding carbon emissions C_i are calculated as:

$$C_i=E_i \cdot \gamma \quad (5)$$

where $\gamma$is the electricity emission factor expressed in kilograms of CO_2 per kilowatt-hour ($kgCO_2/kWh$). The value of γ depends on the energy mix of the assumed electricity grid and can be configured to represent global averages or region-specific scenarios, consistent with prior studies on blockchain carbon emissions [3], [15].

Because γ captures the carbon intensity of the underlying grid, the same energy yields very different emissions across regions and times. We therefore treat γ as an experimental variable with three scenarios: a low-carbon grid (γ = 0.05 $kgCO_2e/kWh$), a global-average grid (γ = 0.475), and a high-carbon grid (γ = 0.82); the sensitivity is reported in the Results.

The total carbon footprint of the blockchain network over a simulation run is computed as:

$$C_{total}=\textstyle\sum_{(i=1..N)} C_i=\gamma \sum_{(i=1..N)} E_i \quad (6)$$

where N denotes the total number of participating nodes.

B. Granular Emission Metrics

To enable fine-grained sustainability analysis, carbon emissions are further normalized with respect to blockchain outputs. Carbon emissions per block $C_{(block)}$ and per transaction C_{tx} are defined as:

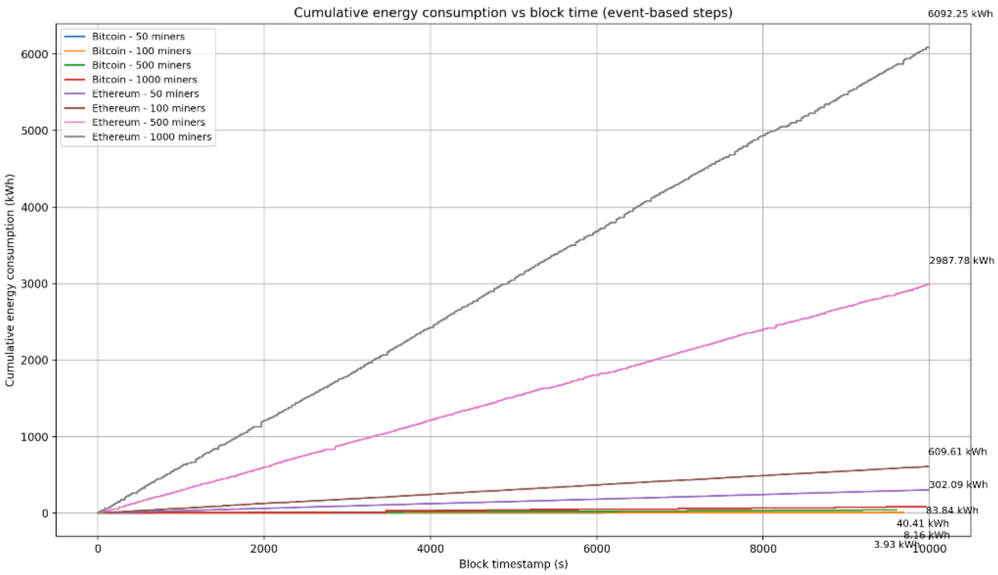
$$C_{(block)}=(C_{total})/(B), C_{tx}=(C_{total})/(T) \quad (7)$$

where B is the total number of confirmed blocks and T is the total number of processed transactions during the simulation. These normalized metrics are commonly used in empirical blockchain sustainability studies to compare protocol efficiency across different workloads and configurations [3], [16].

C. Integration with BlockSim

By embedding carbon footprint computation directly into BlockSim's event-driven execution, the simulator produces time-series and aggregate emission statistics without external post-processing. This tight coupling enables comparative evaluation of blockchain protocols with respect to performance, energy efficiency, and environmental impact within a unified simulation framework.

Experimental Setup



To evaluate the effectiveness of the proposed energy and carbon footprint extensions, a series of controlled simulation experiments were conducted using the extended BlockSim framework. The experimental design follows best practices for blockchain performance evaluation, ensuring reproducibility and comparability with prior studies [4], [16].

A. Network and System Configuration

The simulated blockchain network consists of N nodes, where $N \in \{50,100,500,1000\}$, representing small to moderately sized permissionless networks commonly analyzed in the literature [6]. Nodes are connected via a peer-to-peer overlay with randomly assigned propagation delays drawn from a bounded distribution, consistent with established modeling approaches for blockchain network latency [8]. BlockSim's discrete-event scheduler is used to model message dissemination, block arrival, and chain updates.

B. Consensus and Workload Parameters

The primary evaluation scenario is based on a Proof of Work–style consensus mechanism, which serves as a baseline due to its well-documented energy characteristics [2]. Two PoW-based reference models were evaluated: Bitcoin (Model 1) and a historical Ethereum-like PoW configuration (Model 2). The latter reflects the pre-Merge GPU-mining regime and does not represent current Ethereum, which has used Proof-of-Stake since The Merge. Block generation intervals follow the default targets of each model (Bitcoin: ~10 minutes; Ethereum: ~12–13 seconds), which directly affects the number of block events observed during the same simulation horizon. Transaction workloads are modeled as a Poisson process, a common assumption in blockchain performance studies [4], [17]. To assess scaling effects on sustainability metrics, miner population was varied while maintaining the same simulation time window.

To assess the sensitivity of energy and emission metrics, additional scenarios vary mining power distribution among nodes and transaction throughput. These variations allow for an analysis of how network size and the intensity of the workload affect results related to sustainability.

C. Energy and Carbon Parameters

Energy consumption is computed at runtime using mining-interval accounting (start/stop events). In the evaluated configuration, Bitcoin uses a hashrate-and-efficiency-based power derivation, while Ethereum uses a direct fixed per-miner power configuration (e.g., 2500 W/miner). Carbon emissions are computed using a configurable electricity emission factor; in this revision it is varied across low, average, and high grid scenarios (see Results) rather than held fixed, following established blockchain carbon footprint analyses [3]. Concretely, carbon is derived via the standard emission-factor formulation (electricity use × emissions factor), consistent with common GHG accounting practice.

D. Evaluation Metrics and Methodology

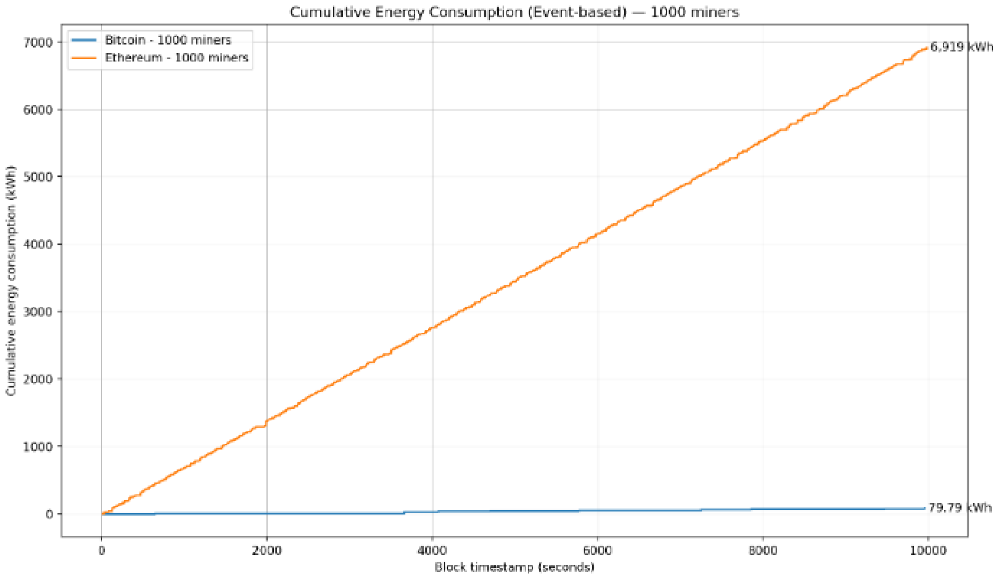
The subsequent metrics are gathered for each simulation scenario: aggregate energy consumption, energy expenditure per block, energy usage per transaction, total CO_2 emissions, and CO_2 emissions per block and transaction. Concurrently, performance metrics, encompassing throughput, propagation delay, and the rate of stale blocks, are documented to facilitate an analysis of the trade-offs inherent in balancing efficiency and sustainability. Each scenario is replicated across 30 independent random seeds (generated deterministically as base + i, i = 0..29), and results are reported as the mean with sample standard deviation and 95% confidence interval (Student t).

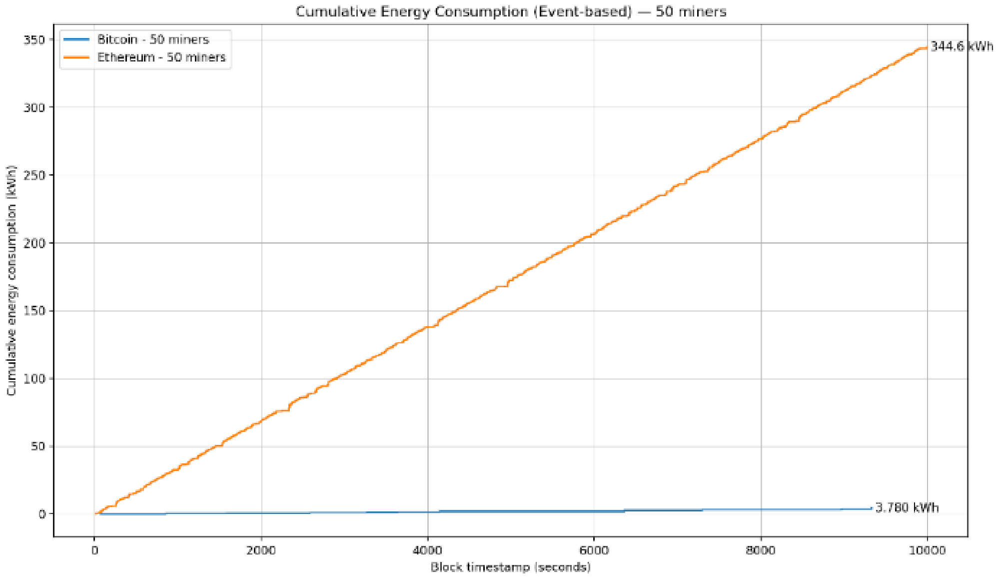
The extended implementation and all code-level modifications to the BlockSim simulator (including the added energy-consumption and carbon-footprint modeling components) are made publicly available in a version-controlled GitHub repository at [18].

Results and Analysis

The simulation results demonstrate that integrating energy and carbon footprint modeling into BlockSim enables meaningful sustainability analysis while preserving the simulator's original performance evaluation capabilities. Across all tested network sizes, total energy and CO_2 increased with network scale, confirming trends reported in prior empirical and simulation-based blockchain studies [4], [16].

However, the magnitude difference between Bitcoin and Ethereum in these runs is dominated by the selected power modeling assumptions, as discussed below.





A. Energy Consumption Analysis

Total energy consumption exhibits strong scaling with miner count in both models. In the reported runs (simulation horizon = 10,000 seconds), the cumulative totals extracted from EnergyLog were:

Bitcoin: 3.93 kWh (50 miners) → 83.84 kWh (1000 miners)

Ethereum: 346.70 kWh (50 miners) → 6934.13 kWh (1000 miners)

This behavior aligns with earlier observations that computational effort scales with participating mining capacity in Proof of Work–based systems [2]. Notably, the Ethereum totals are ~83–88× higher than Bitcoin at the same miner counts in this configuration (e.g., 6934.13/83.84 ≈ 82.7× for 1000 miners). The primary reason is that Ethereum was configured with a fixed per-miner wattage, so total network power grows proportionally with miner count, whereas Bitcoin's effective power is substantially lower under the current parameterization.

These absolute magnitudes are scenario-based model outputs, not real-world measurements, and the Ethereum-labelled curve is a historical Ethereum-like PoW configuration that does not represent current (Proof-of-Stake) Ethereum. Under the added economic PoW model, total network energy is governed by mining economics rather than miner count: holding the economics fixed, the network total is approximately 467.5 GWh over 24 h and is essentially invariant across 50-1000 miners (95% CI ± 17.1 GWh), while mean per-miner energy falls approximately as 1/N. Energy scales linearly with coin price, rising from 155.8 GWh at 20,000 USD to 779.1 GWh at 100,000 USD, and inversely with electricity price, confirming the economic argument of Section D. Figure 7 summarizes the sensitivity of PoW network energy to the number of miners under the revised economic model: the network total is essentially invariant to miner count, while the mean per-miner energy falls approximately as 1/N.

[Insert Figure 7 here: PoW energy sensitivity to miner count]

Figure 7. PoW energy vs number of miners (economic model; 30 seeds, 95% CI). Total network energy is invariant to miner count (a); per-miner energy falls ~1/N (b).

For Proof-of-Stake, the validator-count model yields far smaller totals: a 1000-validator network of standard 100 W servers consumes about 2387 kWh over 24 h, scaling linearly with validator count (239 kWh at 100 validators). The PoW/PoS energy ratio is on the order of 196,846×, consistent with the order-of-magnitude reductions reported in the literature [16], [19] and the ~99.95% reduction at Ethereum's Merge [21]. As an analytical sanity check, fixed-power configurations reproduce the closed-form $E=P \cdot T$ exactly (maximum relative error 0.00e+00). As shown in Figure 8, PoS total energy scales linearly with the number of validators across the low-power, standard-server, and high-power hardware classes.

[Insert Figure 8 here: PoS energy scaling with validator count]

Figure 8. PoS total energy vs validator count for three hardware classes (30 seeds, 95% CI).

Communication energy was also analysed. At peer degree 8 with 1 MB blocks it grows from 15.0 kWh/day at 1 tx/s to 1455 kWh/day at 100 tx/s. Relative to PoW consensus energy this is negligible (ratio ~3e-06), but at high throughput it becomes comparable to PoS consensus energy, so communication energy matters mainly in PoS and high-throughput settings. Figure 9 compares computational (consensus) energy with communication energy under the evaluated scenarios on logarithmic axes.

[Insert Figure 9 here: Computational/consensus energy versus communication energy]

Figure 9. Computation vs communication energy (log-log; peer degree 8, 1 MB blocks, 30 seeds).

Because Ethereum's modeled block interval is much shorter than Bitcoin's, Ethereum produces hundreds of blocks within the same simulation window while Bitcoin produces tens of blocks. Consequently, energy per block is less extreme than total energy (since Ethereum divides its total energy across many more blocks). This effect is visible in the "step density" of event-based curves: Ethereum rises more frequently due to more block events.

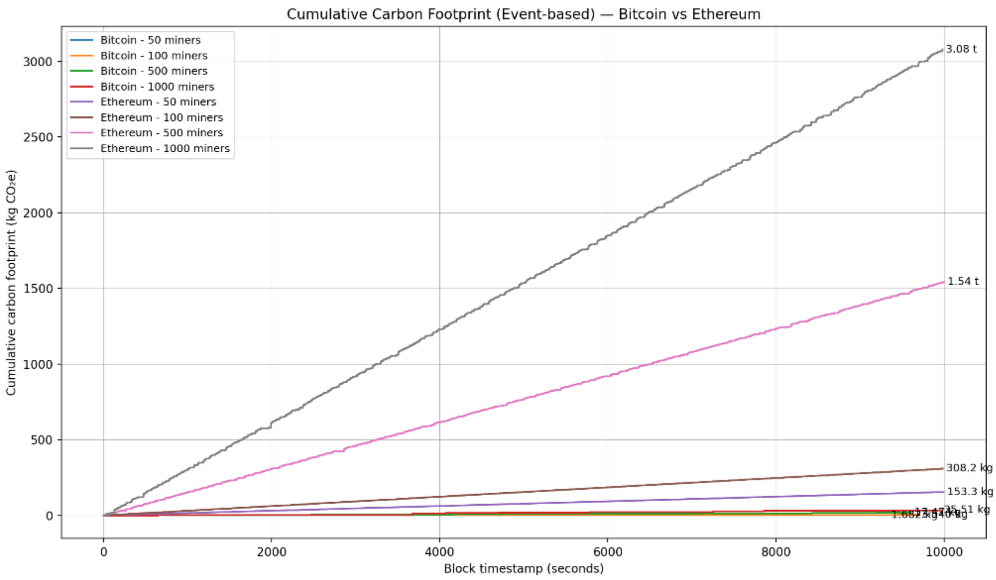


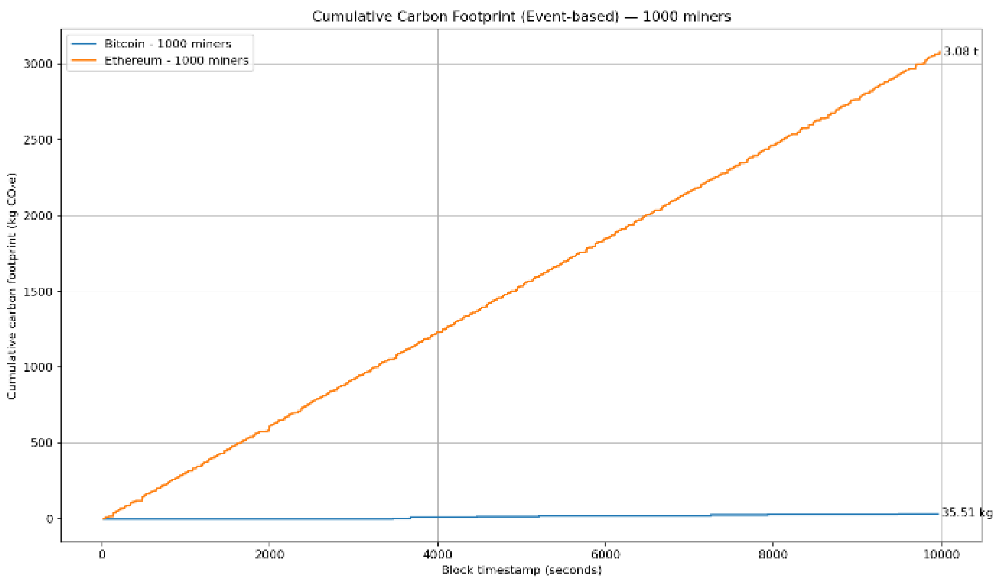
Figure 1. shows the event-based cumulative energy sampled at block timestamps for both models and all miner counts. Ethereum curves rise much more steeply, and scaling with miner count is near-linear.

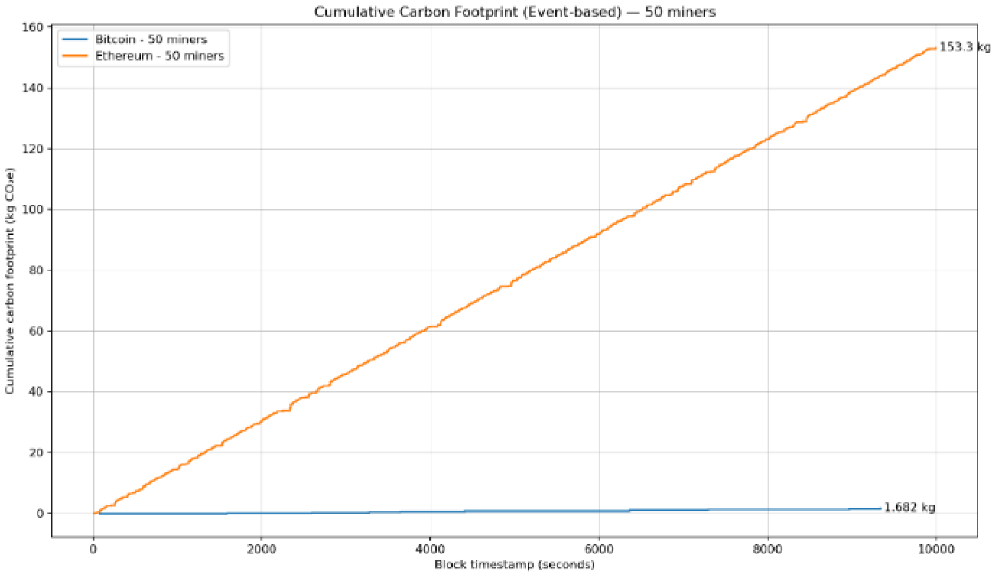
Each figure 2-3 compares Bitcoin vs Ethereum for the same miner count 50 miners and 1000 miners respectively, illustrating that the gap widens predictably with network scale under the current power assumptions.

Overall, these results reinforce the well-known performance-energy relationship in PoW: increasing participating mining capacity increases total energy demand [2], [17]. Additionally, skewed mining power distributions (not shown here) tend to increase aggregate energy because dominant miners perform a larger share of work, consistent with centralization concerns that can worsen efficiency without necessarily improving throughput [8].

B. Carbon Footprint Analysis

Carbon emissions closely follow energy consumption because emissions are computed directly from electricity usage via an emission factor that, in this revision, is varied across low/average/high grid scenarios rather than held fixed [3]. In the evaluated configuration, cumulative CO_2 totals increased proportionally with miner count:





Bitcoin: 1.75 kg CO_2e (50 miners) → 37.31 kg CO_2e (1000 miners)

Ethereum: 154.28 kg CO_2e (50 miners) → 3085.69 kg CO_2e (1000 miners)

As expected, the carbon curves preserve the same ordering and near-linear scaling as energy because the model uses a constant grid factor. This follows standard emissions accounting (kWh × kgCO_2e/kWh).

Varying the grid emission factor shows that geography dominates carbon outcomes for a given energy level. For the PoW reference, daily emissions range from 23,374 t at γ = 0.05 to 383,332 t at γ = 0.82 (about a 16× spread from grid choice alone), while the PoS reference spans only 119-1947 kg. A single fixed emission factor is therefore insufficient. Figure 10 reports the carbon-emission sensitivity under low, average, and high electricity emission factors γ.

[Insert Figure 10 here: Carbon-emission sensitivity under different γ values]

Figure 10. Carbon emissions under low/average/high grid emission factors (log axis; 30 seeds, 95% CI).

Figure. 4 shows cumulative CO_2e (kg) sampled at block timestamps, highlighting the same scaling behavior observed for energy.

Figures 5-6 compares Bitcoin vs Ethereum for the same miner count 50 miners and 1000 miners respectively, illustrating that the gap widens predictably with network scale under the current power assumptions.

By normalizing emissions at the block and transaction levels, the simulator enables direct comparison between protocol configurations, a feature that is often missing in retrospective carbon analyses [3], [15]. These metrics provide actionable insights into how protocol parameters and workloads influence environmental impact.

C. Performance and Sustainability Trade-offs

When performance metrics are analyzed alongside sustainability indicators, a clear trade-off emerges between throughput and environmental efficiency. Scenarios optimized for higher throughput exhibit increased energy consumption and carbon emissions, reinforcing concerns raised in prior work regarding the environmental cost of scaling blockchain systems [2], [17]. The integrated results demonstrate the value of a unified simulation framework that captures performance, energy, and carbon metrics simultaneously.

Overall, the results confirm that the proposed extensions enable BlockSim to serve as an effective tool for sustainability-aware blockchain evaluation, bridging the gap between performance analysis and environmental impact assessment.

Discussion and Threats to Validity

A. Discussion

The findings underscore the necessity of incorporating sustainability metrics directly into blockchain simulation frameworks. This study's extension of BlockSim, incorporating energy consumption and carbon footprint modeling, facilitates a unified assessment of performance, energy efficiency, and environmental impact within a single experimental setting. This integrated approach is crucial, given that previous research has demonstrated that enhancements in throughput and scalability frequently entail increased energy consumption and emissions [4], [17].

A key observation in the Bitcoin vs Ethereum comparison is that energy and CO_2 outcomes are highly sensitive to the chosen power model. In these experiments, Ethereum is configured with a fixed high per-miner power, which causes total network power to scale directly with miner count. This makes Ethereum’s cumulative energy and carbon footprint substantially larger than Bitcoin in absolute terms, even though Ethereum also produces many more blocks during the same simulation window. Methodologically, this emphasizes the need to clearly distinguish between (i) scenarios that represent a growing mining industry (more miners = more total watts) and (ii) scenarios that represent redistribution of a fixed global hashrate across more participants.

The results corroborate that energy and carbon metrics exhibit predictable scaling in relation to network size and workload intensity, consistent with empirical data derived from actual blockchain systems [3]. Furthermore, the capacity to normalize emissions on a per-block and per-transaction basis offers a valuable framework for comparing protocol configurations, thereby mitigating a constraint observed in numerous retrospective sustainability analyses that depend on broad, system-wide estimations [15]. Consequently, this strengthens the efficacy of simulation-based methodologies for investigating "what-if" scenarios during the design phase of blockchain protocols.

Incorporating energy and carbon accounting into BlockSim, from a methodological standpoint, preserves its fundamental modularity and abstraction advantages, simultaneously expanding its applicability to sustainability-oriented research. Consequently, the simulator becomes an essential tool for evaluating innovative consensus mechanisms and optimization approaches aimed at reducing environmental effects, thus augmenting current performance-centered studies [16].

B. Threats to Validity

Several threats to validity should be considered when interpreting the results. Modeling assumptions represent a primary internal validity concern. Energy consumption is approximated via mining-interval accounting and configurable coefficients/power assumptions rather than hardware-specific measurements, which may not fully capture the diversity of real-world mining and validation infrastructures. However, this abstraction is consistent with prior simulation-based energy studies and is necessary for scalable experimentation [9].

The use of a fixed electricity emission factor to calculate carbon emissions limits external validity. In practice, carbon intensity varies significantly across different regions and over time, depending on the energy sources used [3], [15]. Although the model allows for adjustable emission factors, the results presented here reflect average conditions and may not apply to all situations where the model is used.

Construct validity is, in the final analysis, shaped by the degree of abstraction present within BlockSim. Discrete-event simulation, although capable of accurately modeling system-level behavior, is inherently limited in its capacity to fully capture low-level hardware optimizations or the complex nuances of real-world network dynamics. Despite these constraints, this degree of abstraction is a commonly accepted compromise within blockchain research, facilitating systematic comparisons across a variety of protocol designs [4], [16].

Additional limitations follow from the revised scope. First, the framework is a scenario-based explorer; the absolute energy and carbon values are model outputs for the chosen assumptions and are not intended to predict the exact real-world consumption of Bitcoin or Ethereum.

Second, PoW energy is sensitive to coin price, block reward, transaction fees, electricity price, mining efficiency, and difficulty/hashrate dynamics, and to the spend ratio κ; PoS energy is sensitive to validator count, node hardware, uptime, redundancy, and communication overhead.

Third, carbon depends strongly on region- and time-specific grid mixes; the γ scenarios bracket but do not resolve this variability. Finally, historical Ethereum-like PoW results must not be read as current Ethereum, which uses Proof-of-Stake, and communication energy, while small under PoW, can matter in PoS and high-throughput scenarios.

Conclusion and Future Work

This paper presented an extension of the BlockSim simulation framework that integrates energy consumption and carbon footprint modeling into its discrete-event architecture. By embedding mining-interval energy accounting and deriving carbon emissions using established emission-factor–based formulations, the proposed approach enables unified evaluation of blockchain systems in terms of performance, energy efficiency, and environmental impact. Unlike retrospective measurement studies, the extended simulator supports controlled “what-if” experimentation, allowing protocol designers to analyze sustainability implications during the early design phase.

The experimental results demonstrate that energy usage and carbon emissions scale predictably with network size, workload intensity, and consensus configuration, in line with prior empirical and analytical studies of blockchain systems [16], [2]. Furthermore, the capacity to normalize environmental impact on a per-block and per-transaction basis offers a valuable foundation for evaluating different protocol designs, thereby overcoming a significant constraint observed in numerous current blockchain simulation tools [4]. These results corroborate that augmenting BlockSim maintains its advantages in terms of flexibility and abstraction, while simultaneously expanding its utility for research centered on sustainability.

Several avenues for future research are suggested by the findings of this study. Initially, the energy model's accuracy could be enhanced by integrating hardware-specific parameters obtained from empirical data on mining rigs, validator nodes, and networking infrastructure, thereby improving realism while preserving scalability. Furthermore, subsequent developments could incorporate dynamic and region-specific carbon intensity models, allowing for the simulation of diverse energy compositions and renewable energy adoption levels, which have demonstrably influenced blockchain carbon footprints [3]. Furthermore, the framework's scope could be broadened to incorporate emerging low-energy consensus mechanisms, thus facilitating direct comparisons between traditional Proof of Work and alternative, sustainability-oriented designs [12]. Ultimately, integrating the improved BlockSim with more advanced network simulators or real-world data offers a potentially effective approach to reconcile the gap between theoretical modeling and practical blockchain applications.

Overall, this work contributes a practical and extensible foundation for sustainability-aware blockchain simulation, supporting future research on environmentally responsible decentralized systems.

Declarations

Ethics approval: Not applicable. This study is based solely on simulation and does not involve human participants, human data, human tissue, or animals; therefore, ethics committee/IRB approval was not required.

Consent to participate: Not applicable.

Consent for publication: Not applicable. This manuscript does not contain any individual person’s data in any form (including identifiable details, images, or videos).

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